

# GUARDIAN WATCH

## Comprehensive Technical Project Report

### Integrated Wearable Fitness & Health Monitoring Platform

Wearable Hardware • Embedded Firmware • Flutter Mobile App • Firebase / Google Cloud • FastAPI •  
React Web Platform • Manufacturing & Regulatory Roadmap

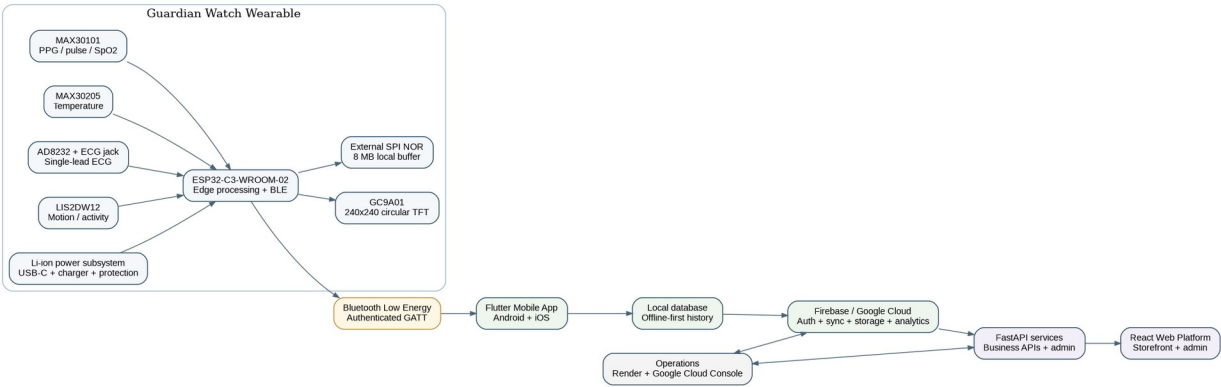


Figure 1. Guardian Watch end-to-end system architecture.

Prepared as an engineering design, research, and commercialization reference

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Nigeria-focused launch strategy with international technical alignment

## Document Control

| Item                     | Value                                                                                                                                                                                                    |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project                  | Guardian Watch                                                                                                                                                                                           |
| Document type            | Comprehensive technical project report                                                                                                                                                                   |
| Primary domains          | Embedded systems, wearable sensing, mobile software, cloud/backend, web commerce, cybersecurity, manufacturing                                                                                           |
| Target market            | Fitness/wellness initially; regulated health-monitoring features subject to validation and regulatory classification                                                                                     |
| Primary geography        | Nigeria; architecture designed for international expansion                                                                                                                                               |
| Report date              | 26 August 2026                                                                                                                                                                                           |
| Status                   | Engineering baseline / pre-production design                                                                                                                                                             |
| Important interpretation | Component selection and architecture are design recommendations; final schematic values, PCB dimensions, clinical claims, certification scope and production test limits require engineering validation. |

**SCOPE AND SAFETY:** This report deliberately distinguishes measured physiological signals, wellness analytics and regulated medical-device claims. A device may use clinical-grade components without itself being a clinically validated medical device. Product claims, algorithm validation, intended use and regulatory classification must be aligned before commercial release.

## Executive Summary

Guardian Watch is proposed as a vertically integrated wearable technology platform that combines a compact physiological sensing device, a cross-platform mobile application and a cloud-connected web/operations environment. The product is designed to capture heart-related and body-state signals at the edge, stream telemetry using Bluetooth Low Energy (BLE), preserve data offline, and expose historical and analytical views through software.

The hardware baseline uses an ESP32-C3-WROOM-02 as the central MCU and BLE controller, a MAX30101 optical sensor for PPG/heart-rate/SpO<sub>2</sub>-related acquisition, a MAX30205 temperature sensor, an AD8232 single-lead ECG front end with an external electrode/jack interface, a low-power 3-axis accelerometer for motion and activity context, a GC9A01 circular display, local SPI flash, rechargeable Li-ion power, USB Type-C charging, protection circuitry and production-test access. The MAX30101 requires a 1.8 V logic supply and a separate LED rail; this has a direct effect on the power architecture and is explicitly addressed in the design. [1][2][3]

The software architecture is intentionally offline-first. The Flutter app handles authentication, Bluetooth permissions and device verification, real-time visualization, local persistence, history, insights, premium entitlements and cloud synchronization. Firebase Authentication provides email/password and Google sign-in and supports password-reset email workflows. [8][9] The web platform uses React for the storefront and administration experience, FastAPI for application APIs, Firebase/Google Cloud services for identity and cloud data, and Render for selected application hosting.

The literature review shows a strong evidence base for wrist-worn heart-rate measurement during rest and common aerobic activities, but error increases during some activities and physiological estimates depend on sensor placement and signal quality. Reviews also identify concerns about heterogeneous validation methods and possible performance differences across skin tones. [10][11][12] Wearable AFib detection and sleep-apnea research is promising, but these features require clinical-grade validation, careful algorithm design and regulatory review; PPG should not be treated as a substitute for confirmatory ECG diagnosis. [13][14][15]

The report therefore recommends a staged product strategy: Phase 1 establishes reliable fitness/wellness telemetry; Phase 2 adds validated trend and fatigue models; Phase 3 evaluates AFib and sleep-apnea algorithms within an explicit clinical/regulatory program. This separation reduces technical debt and protects the product from premature medical claims.

## Abstract

This report presents the research, systems engineering and commercialization design for Guardian Watch, an integrated wearable fitness and health-data platform. The work spans requirements engineering, literature review, sensor selection, embedded hardware, power management, PCB design, firmware architecture, BLE protocol design, mobile and web software, data architecture, cybersecurity, signal processing, AI-assisted insights, manufacturing, testing and Nigerian regulatory planning. The proposed device combines optical photoplethysmography, single-lead electrocardiography, skin/body-temperature sensing and inertial motion sensing with an ESP32-class controller and circular color display. The companion Flutter application provides authenticated account management, secure device onboarding, live telemetry, historical data, insight services and optional cloud synchronization. The web platform supports product discovery, order handling, delivery operations and administration using React and FastAPI. The report integrates published research indicating that wearables can provide useful physiological measurements but that real-world accuracy is influenced by motion, activity type, sensor placement, skin tone, algorithm design and validation quality. The proposed design therefore prioritizes signal-quality gating, offline-first data handling, privacy-by-design, explicit risk management and phased validation. The manufacturing strategy uses PCBWay for PCB fabrication/assembly and includes production test, documentation, packaging, local conformity assessment and regulatory classification workstreams. The resulting architecture is intended to serve both as a build guide for developers and engineers and as a defensible technical foundation for stakeholders, investors, manufacturing partners and future regulatory submissions.

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# 1. Introduction and Background

## 1.1 Context

Wearable computing has evolved from step counters into systems capable of collecting continuous streams of physiological and behavioral data. This shift is enabled by low-power optical sensors, miniature analog front ends, MEMS motion sensors, embedded processors and ubiquitous smartphone connectivity. Modern platforms can capture heart rate, pulse-wave information, temperature, motion and, in some products, single-lead ECG waveforms. The engineering challenge is no longer simply sensor acquisition; it is the reliable integration of sensing, signal quality assessment, context, data storage, user experience, security, algorithm validation and product manufacturing.

Guardian Watch is positioned within this evolution as an engineered platform rather than a single gadget. The device is expected to work as a wearable even when a smartphone is absent for short periods, while the mobile application acts as the primary long-term data interface and optional gateway to cloud services. The design also anticipates a web-based commercial and operational layer for product ordering, delivery management, support and administration.

## 1.2 Technology Context

The selected ESP32-C3-WROOM-02 family combines a 32-bit RISC-V single-core processor with integrated 2.4 GHz Wi-Fi and Bluetooth 5 LE, up to 15 GPIOs and common interfaces such as SPI, I2C and UART. Espressif documents operating supply at approximately 3.0-3.6 V and explicitly lists healthcare, wearable and low-power data-logger use cases among the target applications. [1]

The MAX30101 is a compact optical module integrating LEDs, photodetection, optical elements and ambient-light rejection. It communicates over I2C and is designed for wearable heart-rate and pulse-oximetry applications. Its architecture is unusually relevant to Guardian Watch because the device has separate logic and LED power domains, which require deliberate PCB and power-tree design. [2] The MAX30205 offers a 16-bit digital temperature output and is specified for high accuracy around normal body-temperature range; its operating supply is 2.7-3.3 V. [3]

The AD8232 is a single-lead ECG front end with integrated signal conditioning, configurable high-pass and low-pass filtering, lead-off detection and a low typical supply current. Its intended application set includes portable ECG and remote health monitoring, making it a suitable front-end building block for a prototype while leaving the burden of complete device safety and clinical validation to the overall system. [4]

## 1.3 Why Guardian Watch

The project addresses a practical gap between inexpensive generic fitness bands and high-cost medical monitoring solutions. The design goal is not to replicate every function of a commercial premium smartwatch. Instead, it concentrates engineering effort on a small set of physiological signals and on the surrounding software that makes those signals useful: trustworthy telemetry, session management, history, privacy, device identity, cloud synchronization and insight generation.

# 2. Problem Statement

Many wearable products present physiological metrics without exposing the engineering assumptions, quality indicators or data lineage behind those values. A user may see a heart-rate number without knowing whether the sensor had stable skin contact, whether motion corrupted the optical waveform, whether the reading was a direct measurement or a derived estimate, or whether the record has been synchronized and retained safely. At the product-development level, fragmented hardware and software components can also make manufacturing, debugging and future feature expansion difficult.

Guardian Watch addresses these problems through an integrated architecture in which sensor acquisition, signal-quality checks, user sessions, BLE transport, local storage, cloud synchronization and insights are designed together. It also recognizes that health analytics have a higher burden of proof than general wellness features. For example, recent systematic reviews find strong promise for wearable AFib detection, but also point to false positives, motion artifacts, adherence issues and the need for ECG confirmation and clinical integration. [13][14]

## 3. Aim, Objectives and Research Questions

### 3.1 Aim

To design and specify a professional, manufacturable wearable fitness and health-monitoring platform that integrates physiological sensing, embedded processing, BLE connectivity, mobile software, cloud/backend services and a commercial web platform within a secure and validation-oriented architecture.

### 3.2 Specific Objectives

36. Design a compact wearable electronics platform using the ESP32-C3-WROOM-02, MAX30101, MAX30205, AD8232, inertial motion sensor and GC9A01 display.
37. Develop a safe rechargeable power system with USB Type-C input, Li-ion charging, power-path management, battery protection and fuel-gauge estimation.
38. Design a production-oriented PCB with controlled analog/digital partitioning, test points, ESD protection, decoupling and manufacturability considerations.
39. Define firmware architecture for sensor scheduling, signal quality, power management, BLE telemetry, local buffering and secure device identity.
40. Define a Flutter mobile application for Android and iOS with Firebase authentication, BLE onboarding, real-time dashboard, history, insights, local data storage and cloud synchronization.
41. Define a React/FastAPI web platform for product presentation, ordering, delivery operations, administration and backend management.
42. Establish a research-informed signal-processing and AI/ML roadmap for heart rate, SpO2 estimation, fatigue monitoring, AFib screening and sleep-apnea research.
43. Define manufacturing, testing, quality, documentation, packaging, IP and regulatory workstreams for Nigerian commercialization and future international expansion.
44. Produce a complete engineering baseline bill of materials and software/cloud component inventory suitable for procurement and design review.

### 3.3 Research Questions

45. Which sensor and embedded architecture provides the best balance between physiological measurement quality, power consumption, size and manufacturability?
46. How should wearable sensor data be quality-gated and synchronized to reduce the impact of motion and intermittent connectivity?
47. Which mobile and cloud architecture provides a reliable offline-first experience while preserving data security and user control?
48. What evidence supports wearable-based heart-rate, SpO2, temperature, AFib, sleep-apnea and fatigue monitoring, and what limitations must be reflected in Guardian Watch claims?
49. What hardware, software, manufacturing, cybersecurity and regulatory controls are necessary to move from prototype to a commercial Nigerian product?

## 4. Scope, Assumptions and Constraints

### 4.1 In Scope

- Wearable PCB, sensors, power subsystem, display, buzzer, ECG electrode/jack interface, local storage, enclosure and strap.
- ESP32 firmware, BLE GATT protocol, local buffering, device identity and basic signal processing.
- Flutter Android/iOS application, Firebase authentication, device onboarding, dashboard, history, insights, local database, cloud synchronization and in-app purchases.
- React storefront/admin front end, FastAPI backend, Firebase/Google Cloud integration, deployment and operational monitoring.
- Manufacturing BOM, PCBWay workflow, production testing, packaging, documentation, certification roadmap and risk controls.

### 4.2 Out of Scope for the Initial MVP

- Continuous diagnostic claims for disease detection without clinical validation.
- Blood-pressure measurement as a product claim; the baseline hardware has no validated cuffless BP method.
- Clinical-grade sleep staging; the initial sensor set is insufficient for full polysomnography-equivalent sleep measurement.
- Independent emergency response or ambulance dispatch services.
- Large-scale hospital information-system integration.

### 4.3 Core Assumptions

- The target wearable is a wrist-worn consumer device with an approximate 40-45 mm mechanical footprint.
- The user is expected to own an Android or iOS smartphone for full historical and insight functionality.
- Internet access is intermittent for many users; the core monitoring path therefore remains functional offline.
- Commercialization in Nigeria will use a formal regulatory classification and product-claim review before launch.

## 5. Research Methodology

The report follows a systems-engineering methodology combining literature review, official manufacturer documentation, regulatory source review, requirements analysis and engineering design synthesis. Current official or primary sources were prioritized for component specifications, software platform capabilities and Nigerian compliance requirements, while peer-reviewed systematic reviews were prioritized for clinical/physiological evidence.

| Research stream | Methods                                                                                                   | Primary output                                     |
|-----------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Literature      | Systematic reviews and recent peer-reviewed studies on wearables, PPG, ECG, AFib, sleep apnea and fatigue | Evidence base and design limitations               |
| Hardware        | Manufacturer datasheets, electrical architecture, power and interface analysis                            | Component selection and schematic baseline         |
| Software        | Official Flutter, Firebase and FastAPI documentation                                                      | Implementation architecture and service boundaries |
| Security        | OWASP mobile guidance, BLE qualification documentation, cloud security principles                         | Threat model and security controls                 |
| Regulatory      | SON, NAFDAC, NDPC and international standards sources                                                     | Compliance and validation roadmap                  |
| Manufacturing   | PCB/assembly requirements, BOM discipline and production-test planning                                    | Pilot-to-production workflow                       |

## 6. Literature Review

### 6.1 Wearable Health Monitoring

A living umbrella review of systematic reviews reports that consumer wearables can achieve useful heart-rate measurement performance, but results vary by outcome and test context. The review reports mean heart-rate bias around  $\pm 3\%$  across the evidence base, while SpO<sub>2</sub> error and sleep measurement accuracy show different patterns and larger uncertainty. It concludes that heterogeneous validation methods make universal accuracy statements difficult and calls for more standardized protocols. [12] This strongly supports Guardian Watch using signal-quality indicators, explicit measurement contexts and validation datasets rather than treating all readings as equally reliable.

### 6.2 PPG and Heart Rate

A meta-analysis of wrist-worn photoplethysmography devices found small mean differences versus ECG or chest-strap references during sleep, rest and treadmill activities, but larger errors during resistance training and cycling. [10] This distinction matters because movement-induced artifact, grip changes and optical contact can distort PPG waveforms even when the underlying sensor is high quality.

Another systematic review examining skin tone found the evidence inconclusive overall, but some included studies reported reduced heart-rate accuracy in darker-skinned participants while others did not observe a significant effect. [11] A professional design should therefore include diverse validation cohorts and should not assume that a calibration tested only on a narrow population generalizes to all users.

### 6.3 ECG and AFib

Wearable ECG research increasingly supports the use of single-lead systems as convenient screening tools for rhythm abnormalities. A recent systematic review of wearable devices reports high performance for wearable ECG patches and moderate-to-high performance for pulse-irregularity detection using PPG, while emphasizing motion artifacts, false positives and the need for confirmatory clinical ECG for abnormal screening results. [13] Another 2025 systematic review of wearable ECG devices found high reported sensitivity and specificity ranges in the included studies, particularly under supervised conditions, but still identified inconclusive recordings and real-world validation as barriers. [14]

**AFIB EVIDENCE:** Design implication: Guardian Watch should implement an AFib research pipeline as a screening/decision-support function first. The product should not market “AFib diagnosis” merely because a classifier performs well on a laboratory dataset. Intended use, ground truth, participant diversity, prospective validation, false-positive burden, clinical workflow and regulator expectations must all be addressed.

### 6.4 Sleep Apnea Monitoring

Recent systematic reviews show significant progress in using wearable sensors and AI for sleep-apnea detection. A 2025 review highlights PPG, motion and other wearable signals together with preprocessing and AI methods; another 2025 meta-analysis evaluates wearable/contactless technologies relative to polysomnography. [15][16] These findings justify research into sleep-related insights but do not imply that a wrist wearable with PPG and pulse oximetry alone can immediately replace a clinical sleep study.

### 6.5 Fatigue Monitoring

Fatigue monitoring is particularly suitable for a multimodal architecture. A 2025 systematic review describes wearable fatigue systems that combine ECG, PPG, motion and other physiological signals with machine

learning. Earlier state-of-the-art work found 60 relevant wearable studies but warned that many were short laboratory investigations with weak generalizability and insufficient attention to signal quality. [17][18] A recent review of HRV-based fatigue detection also reports large variation in field performance and calls for real-world testing. [19]

For Guardian Watch, a fatigue score should therefore be framed as a longitudinal wellness estimate using personal baselines rather than as a binary medical diagnosis. The model should incorporate data quality, recent activity load and user context, and it should expose uncertainty rather than presenting a single unexplained label.

## 6.6 Temperature and Fever/Infection Insights

The MAX30205 is specified as a human-body-temperature sensor with high resolution and an accuracy specification around normal body-temperature range. [3] However, wrist/skin temperature is not identical to core temperature. Temperature estimates can depend on contact pressure, ambient conditions, skin perfusion, strap fit and device self-heating. Therefore, the product should describe the initial fever/infection feature as a trend/screening signal and require a validated measurement protocol before making diagnostic claims.

## 6.7 Literature Review Synthesis

| Domain          | Evidence summary                                                     | Guardian Watch implication                                              |
|-----------------|----------------------------------------------------------------------|-------------------------------------------------------------------------|
| Wrist PPG HR    | Strong evidence of useful HR measurement; activity-dependent error   | Use motion/contact quality gating; validate across activity types       |
| SpO2            | Useful but sensitive to fit, motion, skin optics and calibration     | Use validated calibration curve, reject low-quality windows             |
| Single-lead ECG | Strong potential for rhythm screening                                | Use ECG as a confirmatory modality and preserve raw waveform provenance |
| AFib            | High research potential; screening is more defensible than diagnosis | Clinical validation + regulator review required                         |
| Sleep apnea     | Promising wearable/AI field but PSG remains reference                | Treat premium feature as research/clinical-validation workstream        |
| Fatigue         | Multimodal wearables show promise; field generalization is weak      | Use personal baselines + quality gates + longitudinal validation        |
| Temperature     | Sensor can be accurate but wrist location is context-dependent       | Use trend monitoring and calibration; do not equate to core temperature |

# 7. Competitive and Technology Landscape

The competitive landscape can be divided into four broad categories: low-cost fitness bands, premium smartwatches, specialized ECG/wellness devices and clinical monitoring systems. Guardian Watch should not attempt to compete on app-store ecosystem breadth alone. Its differentiation should come from an integrated engineering proposition: focused physiological sensing, transparent data lineage, offline-first operation, local manufacturing readiness, and a roadmap from wellness telemetry toward validated health insights.

| Category                | Typical strength               | Typical weakness                                  | Guardian opportunity                                                          |
|-------------------------|--------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------|
| Entry fitness trackers  | Low cost; activity tracking    | Limited transparency and physiological validation | Offer stronger signal-quality and data controls                               |
| Premium smartwatches    | Excellent UX; large ecosystems | Higher cost; closed ecosystems                    | Focus on specialized health workflows and local support                       |
| Specialized ECG devices | High-quality ECG workflows     | Less continuous; limited activity context         | Combine ECG session quality with continuous wearable context                  |
| Clinical monitors       | High clinical performance      | Cost, comfort and clinical workflow burden        | Build a comfortable screening/wellness front end, not a premature replacement |

## 8. System Requirements Specification

### 8.1 Functional Requirements

| ID    | Requirement                                                            | Layer           |
|-------|------------------------------------------------------------------------|-----------------|
| FR-01 | Acquire PPG signal and derive heart-rate and SpO2-related features     | Wearable        |
| FR-02 | Acquire skin/body-temperature measurements                             | Wearable        |
| FR-03 | Acquire single-lead ECG through an external electrode/jack session     | Wearable        |
| FR-04 | Acquire 3-axis motion data for activity and motion-artifact assessment | Wearable        |
| FR-05 | Display live values, states and alerts on GC9A01                       | Wearable        |
| FR-06 | Stream telemetry over BLE using versioned GATT services                | Wearable/mobile |
| FR-07 | Buffer telemetry locally when the phone is disconnected                | Wearable        |
| FR-08 | Authenticate users with Firebase email/password and Google OAuth       | Mobile          |
| FR-09 | Support password-reset email workflow                                  | Mobile          |
| FR-10 | Request and manage platform Bluetooth permissions                      | Mobile          |
| FR-11 | Verify a device using identity and authenticated protocol              | Mobile/wearable |
| FR-12 | Display live dashboard and history with date-range filters             | Mobile          |
| FR-13 | Provide free wellness insights and premium entitlements                | Mobile/cloud    |
| FR-14 | Store data locally and synchronize to Firebase/Google Cloud            | Mobile/cloud    |
| FR-15 | Provide profile, theme, photo and data-management settings             | Mobile          |
| FR-16 | Provide public React product storefront and ordering workflow          | Web             |
| FR-17 | Provide admin panel and operational APIs                               | Web/backend     |
| FR-18 | Provide order/delivery states and audit trails                         | Web/backend     |
| FR-19 | Provide logs, monitoring and error reporting                           | Cloud/ops       |

### 8.2 Non-Functional Requirements

| ID     | Quality attribute | Requirement                                                                                                        |
|--------|-------------------|--------------------------------------------------------------------------------------------------------------------|
| NFR-01 | Safety            | Charging, battery, electrical and mechanical safety controls                                                       |
| NFR-02 | Privacy           | Data minimization, consent, access/deletion/export controls                                                        |
| NFR-03 | Security          | Authenticated BLE, secure cloud identity, least privilege, encrypted transport                                     |
| NFR-04 | Availability      | Offline monitoring and local persistence; resilient cloud synchronization                                          |
| NFR-05 | Performance       | Live dashboard update target 1-5 Hz for user-facing values, higher-rate ECG streaming in session mode              |
| NFR-06 | Power             | Display duty cycling and sensor scheduling to target approximately one-day or more battery life depending on usage |
| NFR-07 | Maintainability   | Versioned firmware, APIs, schemas and hardware revisions                                                           |
| NFR-08 | Manufacturability | Standard SMT components, test points, panelization readiness and controlled BOM                                    |
| NFR-09 | Accessibility     | Readable UI, clear alert semantics, theme support and non-color-only status indicators                             |
| NFR-10 | Traceability      | Unique device identifier, firmware version, PCB revision and data provenance                                       |

## 9. Overall System Design

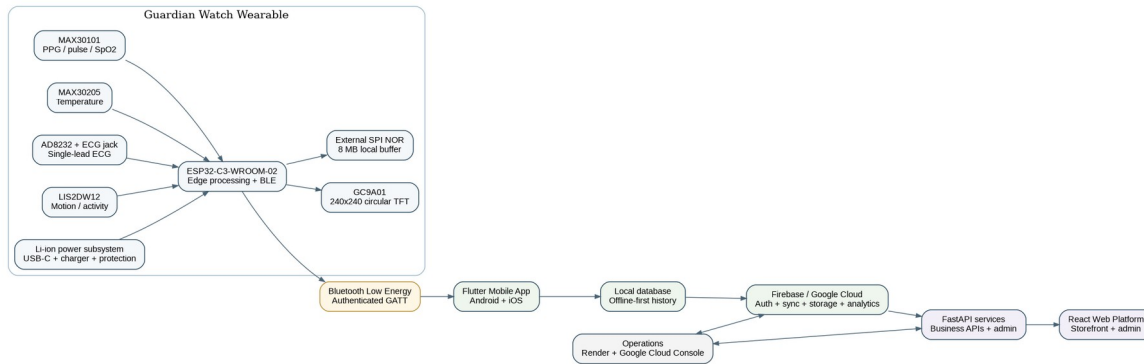
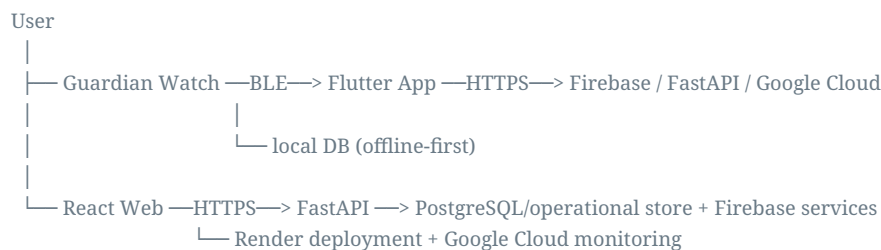


Figure 2. High-level architecture from sensing hardware to cloud/web operations.

### 9.1 Architecture Principles

- **Modularity:** hardware drivers, signal processing, BLE protocol and application features are separated behind explicit interfaces.
- **Offline-first operation:** measurement and local persistence do not depend on internet access.
- **Quality-aware telemetry:** readings carry quality/context metadata so the application can distinguish stable data from low-confidence windows.
- **Secure identity:** device association is stronger than a visible serial number and uses an authenticated handshake.
- **Cloud optionality:** cloud sync improves backup and cross-device access but does not become a single point of failure for basic monitoring.
- **Validation-first health features:** algorithms are developed as independently testable modules with datasets, versioning and performance thresholds.
- **Manufacturing traceability:** each device carries hardware revision, firmware version and unique device identity.

### 9.2 Deployment Topology



## 10. Wearable Hardware Engineering Design

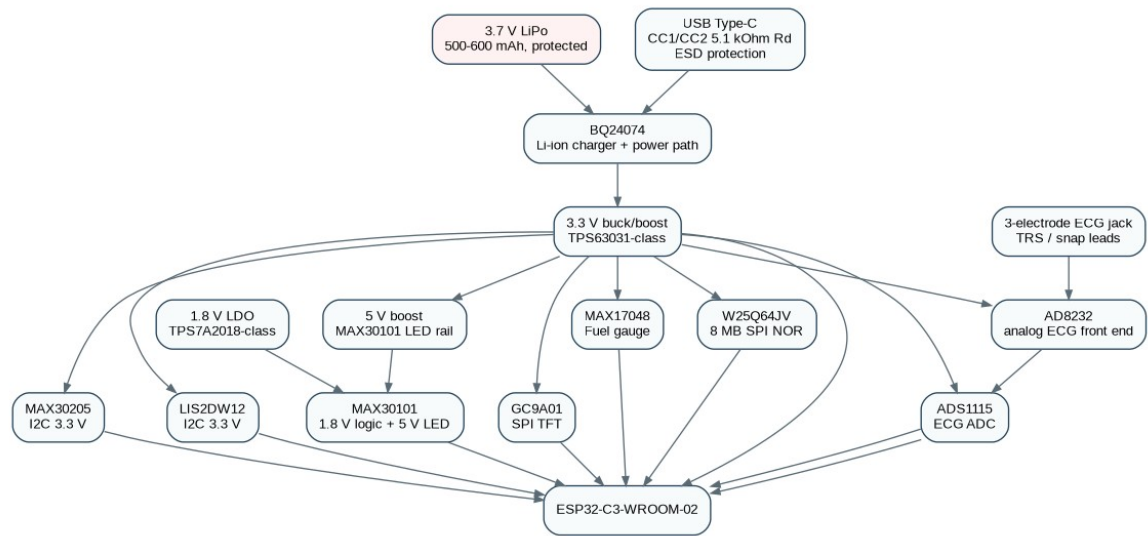


Figure 3. Proposed wearable hardware block diagram and power domains.

### 10.1 Core Controller

The recommended MCU/module is ESP32-C3-WROOM-02 because the product needs BLE, multiple I2C/SPI peripherals, a compact module, low-power modes and sufficient local compute for sensor scheduling and lightweight feature extraction. The module integrates flash and a 40 MHz crystal and provides a PCB-antenna option. [1]

| Subsystem      | Interface              | Power domain               | Engineering note                                                        |
|----------------|------------------------|----------------------------|-------------------------------------------------------------------------|
| MAX30101       | I2C + interrupt        | 1.8 V logic + 5 V LED rail | Requires level shifting and controlled optical/mechanical cavity        |
| MAX30205       | I2C + alert optional   | 3.3 V                      | Place near skin-contact thermal path but thermally isolate from hot ICs |
| LIS2DW12       | I2C/SPI                | 3.3 V                      | Used for motion/activity and PPG motion-artifact gating                 |
| AD8232         | Analog out + leads-off | 3.3 V                      | Keep ECG analog routing short and away from DC/DC switch nodes          |
| ADS1115        | I2C                    | 3.3 V                      | External ADC for more controlled ECG digitization                       |
| GC9A01         | 4-wire SPI             | 3.3 V module logic         | Backlight should be PWM or load-switched                                |
| External flash | SPI                    | 3.3 V                      | 8 MB circular buffer for telemetry and ECG sessions                     |
| MAX17048       | I2C                    | Battery/System             | Fuel-gauge estimate and low-battery management                          |

### 10.2 Why Add an Accelerometer

A 3-axis accelerometer is not merely a “fitness” extra. It is a measurement-quality instrument. The LIS2DW12 operates at very low power, supports 1.6-1600 Hz output rates and includes a FIFO and motion-related functions. [20] In Guardian Watch it is used to identify high-motion windows, estimate activity intensity and gate optical/HRV-derived insights that would otherwise be contaminated by movement.

### 10.3 PPG Subsystem

The optical subsystem uses MAX30101 with an isolated optical cavity, black/opaque barriers to reduce cross-talk and ambient-light leakage, controlled skin pressure and a mechanical alignment target. The sensor is located on the underside of the watch where it remains in stable contact with the wrist. The PCB should keep the optical module away from high-current backlight and RF antenna regions. MAX30101 specifications identify a 5.6 x 3.3 x 1.55 mm package, I2C interface and programmable sample/LED currents. [2]

### 10.4 Temperature Subsystem

The MAX30205 provides a digital temperature value and high nominal resolution. [3] The mechanical design should place the device behind a thermally conductive contact region while avoiding direct coupling to the display, regulator, charging IC and ESP32. During charging, the app should flag temperature measurements as potentially biased by self-heating.

### 10.5 ECG Subsystem

The AD8232 is used as the analog front end for a single-lead ECG session. [4] The ECG jack should connect to a controlled electrode cable or snap-electrode harness. A session should require explicit user initiation, clear electrode placement guidance and a signal-quality check before recording. The PCB should separate analog ground strategy from noisy switching-regulator return currents and should provide a clean reference/bias path for the AD8232.

### 10.6 Display and Interaction

The GC9A01 family supports 240x240 color TFT displays over SPI and includes display RAM and power-saving modes. [5] The selected circular module should be treated as a complete electromechanical part with a defined active area, FPC orientation, backlight characteristics and supply rails. Display brightness should be controlled by PWM/load switching and should default to a low duty cycle to preserve battery life.

## 11. PCB and Electrical Design

### 11.1 Layer Stack Recommendation

A 4-layer PCB is recommended for the production revision: Layer 1 signal/components, Layer 2 near-solid ground, Layer 3 power distribution and controlled digital/analog routing, Layer 4 signals. A 2-layer prototype may be possible, but a 4-layer stack materially improves RF ground integrity, analog return paths and EMI management.

| Layer | Primary purpose                                              |
|-------|--------------------------------------------------------------|
| L1    | Components, short sensitive signals, sensor interfaces       |
| L2    | Continuous ground plane; controlled return path              |
| L3    | 3.3 V, 1.8 V, 5 V boost output, battery/system distribution  |
| L4    | Digital routing, low-speed signals and secondary connections |

### 11.2 Layout Zones

- RF zone: ESP32 antenna end kept clear of copper/components according to Espressif layout guidance. [1]
- Optical zone: MAX30101 underside cavity and local ground/optical barrier, isolated from bright LEDs and switching nodes.
- Analog zone: AD8232 + ECG input + ADC, with short traces and controlled analog return.
- Power zone: USB-C, ESD, charger, switching regulators and inductors physically grouped away from ECG/PPG paths.
- Display zone: GC9A01 SPI/backlight routed away from ECG input and temperature sensor.

- Thermal zone: MAX30205 placed away from heat sources and ideally near the skin-contact interface.

### 11.3 Decoupling and Passive Network

Every IC should receive the manufacturer-recommended local decoupling network. As a starting rule, use 100 nF ceramic capacitors at each IC supply pin or supply group plus 1-10  $\mu$ F local bulk capacitance per functional rail. Switching regulators require the exact inductor and capacitor values from their datasheets and must not be substituted casually because stability can depend on component characteristics.

USB Type-C sink detection should use the appropriate CC pull-down implementation; a 5.1 kOhm  $R_d$  on CC1 and CC2 is the standard baseline for a simple USB 2.0 sink design. [21] USB ESD protection should be placed adjacent to the connector with an extremely short return path.

### 11.4 Test Points

| TP   | Signal     | Purpose                      |
|------|------------|------------------------------|
| TP01 | VBAT       | Battery rail inspection      |
| TP02 | VSYS       | System rail                  |
| TP03 | 3V3        | 3.3 V regulator verification |
| TP04 | 1V8        | MAX30101 logic rail          |
| TP05 | 5V_LED     | Optical LED rail             |
| TP06 | GND        | Test reference               |
| TP07 | ECG_OUT    | Analog front-end output      |
| TP08 | SDA/SCL    | I2C diagnostic access        |
| TP09 | UART_TX/RX | Firmware manufacturing log   |
| TP10 | EN/BOOT    | Programming/recovery         |

### 11.5 Design-for-Manufacture

Use 0402/0603 passives where the assembly house and yield data support them; 0603 can be preferable for first-production serviceability. Keep polarized components visibly keyed, minimize unnecessary unique parts, use consistent footprints and maintain an Approved Vendor List (AVL). Add fiducials, panelization rails, tooling holes and a test fixture interface for production.

## 12. Power, Charging and Battery Safety

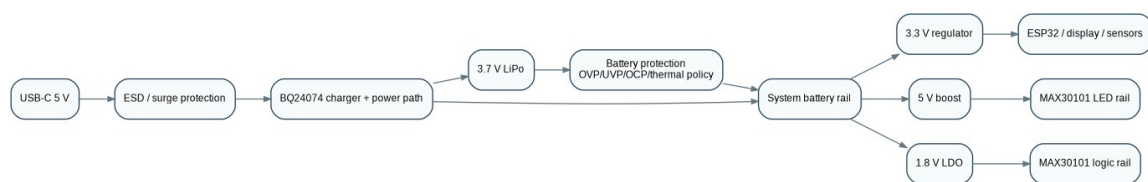


Figure 4. Proposed power tree.

### 12.1 Battery Recommendation

A single-cell 3.7 V nominal Li-polymer battery in the 500-600 mAh range is a reasonable baseline for a 40-45 mm watch enclosure. Final capacity is a mechanical/power trade-off: larger capacity improves endurance but increases thickness, mass and mechanical stress. The battery should be procured from a traceable supplier with cell test documentation and the required transport/test summary.

### 12.2 Charger and Power Path

A professional design should use a charger/power-path IC rather than a bare TP4056 board. The recommended architecture uses a BQ24074-class single-cell charger/power-path device so the system can operate while

charging and can apply controlled current/thermal behavior. The exact production part must be validated for package availability and local assembly capability.

## 12.3 Battery Protection

- Use a protected LiPo pack or implement cell overvoltage, undervoltage, overcurrent and short-circuit protection on the PCB.
- Use temperature-aware charging policy and disable/derate charging when the measured battery/board temperature is outside validated bounds.
- Implement firmware low-battery states: NORMAL, LOW, CRITICAL, SHUTDOWN.
- Provide hardware-level protection that does not depend on firmware executing correctly.
- Require the selected cell/battery assembly to have the evidence needed for relevant transport tests; UN 38.3 addresses design testing of lithium cells and batteries before transport. [22]

## 12.4 Fuel Gauge

MAX17048 is recommended for state-of-charge estimation because it is a low-power single-cell fuel gauge intended for portable equipment and uses a model-based approach rather than a simple resistor divider. [23] The firmware can expose an estimated percentage and low-battery thresholds, while avoiding false precision by smoothing the display value.

## 12.5 Power Budget

| Load                         | Design allowance                                                      | Power strategy                                      |
|------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------|
| ESP32-C3 BLE + processing    | 10-30 mA average depending on duty cycle                              | Use deep/modem sleep and event-driven sensing       |
| GC9A01 backlight + refresh   | 5-40 mA depending on module/brightness                                | PWM brightness and auto-off                         |
| MAX30101                     | sub-mW core in low-power HR mode; LED current dominates session usage | Duty-cycle optical sampling                         |
| MAX30205                     | ~0.6 mA typical active supply current                                 | Use one-shot/periodic conversion                    |
| AD8232                       | ~0.17 mA typical                                                      | Power ECG front end only for sessions when possible |
| LIS2DW12                     | <1 mA in low-power active modes                                       | Use FIFO and low ODR for baseline tracking          |
| Regulators + quiescent loads | 2-8 mA design allowance                                               | Select low-Iq regulators                            |
| Buzzer                       | 10-30 mA momentary, design dependent                                  | Short alerts only                                   |

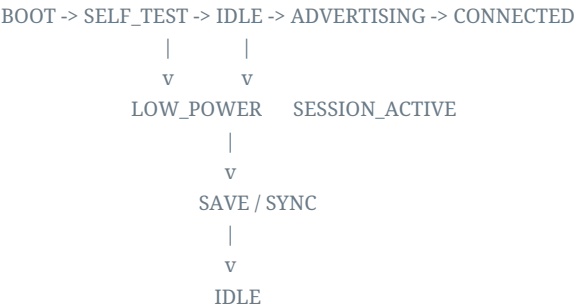
A 550 mAh battery can theoretically deliver 550/average\_mA hours before accounting for usable-capacity limits and conversion losses. If the device averages 25 mA, the idealized runtime is 22 hours; at 15 mA it is about 36.7 hours. These values are engineering estimates, not promises. The production target should be set after measured current profiling in representative user scenarios.

# 13. Embedded Firmware Design

## 13.1 Firmware Architecture

| Module           | Responsibilities                                                          |
|------------------|---------------------------------------------------------------------------|
| Boot/diagnostics | Startup self-test, watchdog, firmware revision, manufacturing diagnostics |
| Power manager    | Battery state, sleep/wake, display backlight, charging state              |
| Sensor manager   | Sensor initialization, scheduling, FIFO handling, sampling profiles       |
| Signal quality   | Motion/contact/noise assessment and validity flags                        |
| Data manager     | Timestamping, framing, compression, local ring buffer                     |
| BLE manager      | Advertising, connection, GATT service handlers, encryption state          |
| UI manager       | GC9A01 screens, alerts, session navigation                                |
| Security manager | Device identity, nonce/challenge handling, secure provisioning            |
| Update manager   | Signed/verified firmware update mechanism in later revisions              |

13.2 State Machine



13.3 Sampling Policies

| Mode           | PPG                          | Temp      | Motion    | ECG                 | Display | Purpose                         |
|----------------|------------------------------|-----------|-----------|---------------------|---------|---------------------------------|
| Idle           | periodic low duty            | 30-60 s   | 5-25 Hz   | off                 | sleep   | Battery preservation            |
| Live dashboard | 25-100 Hz internal           | 5-15 s    | 25-50 Hz  | off                 | on      | Real-time user display          |
| ECG session    | PPG optional                 | 30-60 s   | 25-50 Hz  | 250-360+ SPS target | on      | ECG waveform capture            |
| Fitness        | 25-100 Hz + activity windows | 15-60 s   | 25-100 Hz | off                 | on/dim  | Activity and heart-rate context |
| Sync           | as needed                    | as needed | as needed | buffered            | on/off  | Transfer stored records         |

**ECG SAMPLING:** The selected ECG sample rate is an engineering starting point. The final sample rate, analog bandwidth, anti-aliasing design and reference-device comparison must be locked by the ECG validation protocol.

14. BLE Communication and Device Security

14.1 GATT Service Model

| Service           | Characteristic       | Direction      | Payload concept                                   |
|-------------------|----------------------|----------------|---------------------------------------------------|
| Device Service    | Device ID            | Read           | Public model + non-secret identity                |
| Device Service    | Challenge/Response   | Bi-directional | Authenticated onboarding handshake                |
| Telemetry Service | Vitals               | Notify         | Timestamp, HR, SpO2 estimate, temp, quality flags |
| PPG Service       | PPG samples          | Notify         | Windowed raw/feature data                         |
| ECG Service       | ECG samples          | Notify         | Sequence + sample block + quality                 |
| Sync Service      | Record metadata      | Read/Write     | Start/stop/retry offsets                          |
| Status Service    | Battery/device state | Read/Notify    | Battery, charging, firmware, faults               |
| Control Service   | Session command      | Write          | Start/stop/configuration                          |

14.2 Secure Device Pairing

A visible serial number is not a secret. The secure association should use a device-unique identity and an authenticated challenge/response. A practical flow is: the app discovers the device; the device advertises a non-secret device identifier; the app initiates onboarding; the device returns a random nonce; the app and device derive/verify an authentication result using a provisioned secret or key pair; only then does the device enable the trusted telemetry characteristic.

Provisioning should occur during manufacturing with a secure station or controlled factory process. The production system must protect device secrets from appearing in public logs or firmware source code. BLE security should use encrypted links where supported and should be treated as one layer of security rather than the only layer.

### 14.3 Bluetooth Qualification and RF Compliance

Bluetooth SIG maintains a formal qualification process intended to support interoperability and use of the Bluetooth brand. The qualification flow includes product details, design specification, administrative fee, submission and verification. [24] Because the ESP32 module has an integrated qualified radio, the final end-product compliance strategy still requires assessment of the exact product/radio configuration, market and antenna implementation.

## 15. Mobile Application Design

### 15.1 Platform

Flutter is the recommended cross-platform application framework. The current Flutter supported-platform documentation confirms deployment support for Android and iOS, with current supported OS ranges evolving over time; production builds should pin a tested Flutter SDK release and CI matrix. [25]

### 15.2 Application Layers

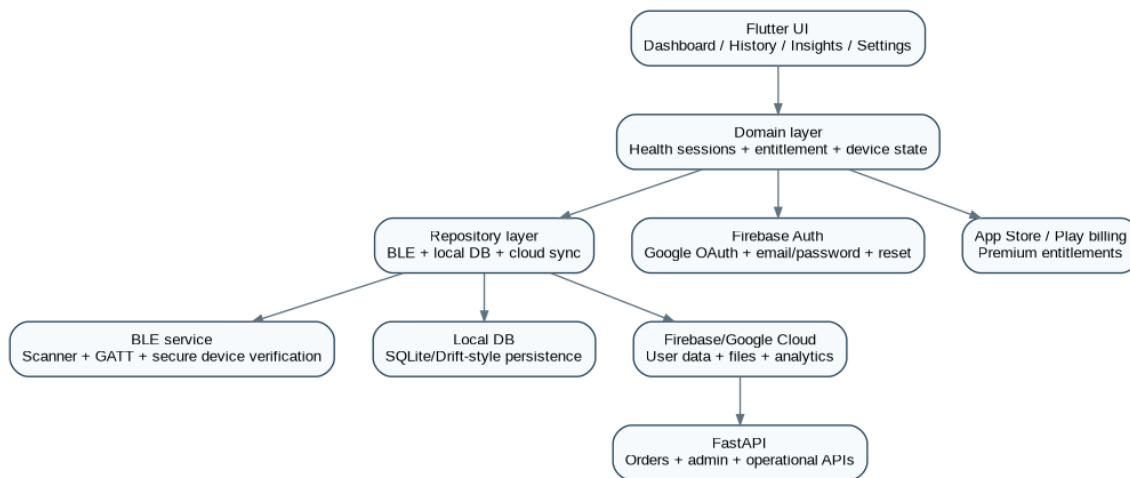


Figure 5. Recommended Flutter application architecture.

### 15.3 Authentication

Firebase Authentication should provide email/password and Google sign-in. Firebase documents both methods and supports password-reset email workflows. [8][9] Authentication state should be managed centrally and never duplicated as a separate password database inside Guardian Watch.

### 15.4 Core Screens

| Screen          | Functionality                                                           |
|-----------------|-------------------------------------------------------------------------|
| Onboarding      | Account creation, privacy notice, permissions, device setup             |
| Login/Register  | Google OAuth, email/password, validation                                |
| Forgot password | Request password-reset email                                            |
| Device Setup    | BLE scan, permission handling, identity verification, connection status |
| Dashboard       | Live HR, pulse, SpO2, temperature, ECG entry point, battery/device      |

| Screen                | Functionality                                                             |
|-----------------------|---------------------------------------------------------------------------|
|                       | state                                                                     |
| ECG Session           | Lead placement, signal quality, live waveform, record/save/share controls |
| History               | Date range selection, trends, session records, export/delete controls     |
| Insights              | Free and premium insight cards, confidence and explanation                |
| Settings              | Profile, photo, theme, device, cloud sync, privacy and data management    |
| Purchase/Entitlements | Store subscription, premium status, restore purchases                     |

## 15.5 Bluetooth Permission Workflow

50. Check whether Bluetooth is available and enabled.
51. Request the platform permissions needed for scanning/connecting under the current OS version.
52. Scan only while the device setup page is active unless a persistent connection feature is explicitly enabled.
53. Present discovered Guardian Watch devices with human-readable name and non-secret device identifier.
54. Perform authenticated identity verification before trusting telemetry.
55. Persist the association as a user-device record and support controlled unlinking/re-pairing.
56. Handle disconnect/reconnect with retry backoff and explicit status feedback.

## 15.6 Dashboard Data Model

| Metric      | Raw source    | Derived value             | Quality metadata                                  |
|-------------|---------------|---------------------------|---------------------------------------------------|
| Heart rate  | PPG           | BPM                       | signal quality, motion state, timestamp           |
| SpO2        | Red/IR PPG    | percentage estimate       | contact/quality state, calibration version        |
| Temperature | MAX30205      | deg C                     | sensor/charging state, calibration version        |
| ECG         | AD8232 + ADC  | waveform/features         | sample rate, lead-off, quality, algorithm version |
| Activity    | Accelerometer | activity class/step proxy | motion confidence, sampling profile               |

# 16. Local Database and Cloud Data Architecture

## 16.1 Data Flow

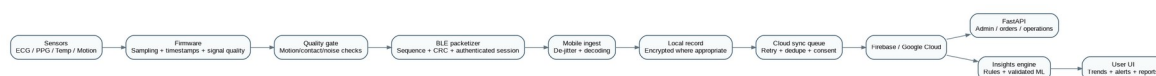


Figure 6. Data pipeline and quality gate.

## 16.2 Local Storage Schema

| Entity      | Key fields                                                | Notes                                              |
|-------------|-----------------------------------------------------------|----------------------------------------------------|
| UserProfile | user_id, display_name, email, photo_ref                   | Firestore user identity; avoid duplicating secrets |
| Device      | device_id, hw_rev, fw_rev, last_seen                      | Association and diagnostics                        |
| Telemetry   | id, device_id, ts, metric, value, quality                 | Normalized or time-series storage                  |
| ECGSession  | session_id, start, end, fs, file_ref                      | Large waveform stored as binary/file chunks        |
| Insight     | id, type, model_version, result, confidence, generated_at | Keep model/version provenance                      |
| SyncQueue   | id, entity, operation, attempts, state                    | Offline synchronization                            |
| Entitlement | product_id, platform, status, valid_until                 | Premium access state                               |

## 16.3 Cloud Strategy

Firebase is recommended for identity and selected user data, while a FastAPI backend provides operational APIs and integration boundaries. Render can host the FastAPI and React services. For production, order/fulfillment data can be stored in a managed PostgreSQL database associated with the backend, while health data is kept in an explicitly permissioned cloud store with strict security rules. This separation prevents e-commerce operations from being tightly coupled to health-record storage.

## 16.4 Synchronization Rules

- Every measurement record receives a device-generated UUID or monotonically increasing sequence identifier plus timestamp.
- Uploads are idempotent; retrying an upload must not create duplicates.
- Cloud synchronization is queued locally and resumed on connectivity.
- Conflicting user settings use last-write-wins only where the data semantics make that safe; health records are append-only except for user-controlled deletion/redaction.
- ECG files are chunked or stored as compressed binary objects, with metadata in the database.
- Users can request deletion and the system must propagate deletion according to the retention policy.

# 17. Signal Processing and Insight Algorithms

## 17.1 General Processing Pipeline

```
Raw samples
  -> timestamp normalization
  -> hardware/lead/contact checks
  -> band-pass / notch filtering
  -> artifact and motion assessment
  -> feature extraction
  -> physiological estimate
  -> confidence score
  -> store / display / insight engine
```

## 17.2 Heart Rate from PPG

A practical baseline uses a band-limited PPG waveform, peak detection with adaptive thresholds and an inter-beat interval plausibility gate. Motion samples from the accelerometer can be used to suppress or down-weight windows with strong periodic interference. The software should retain both the estimate and the quality state.

## 17.3 SpO2

SpO2 estimation should use the red/infrared components of the PPG signal. A standard ratio-of-ratios approach can be used as the starting formulation:  $R = (AC_{red}/DC_{red}) / (AC_{IR}/DC_{IR})$ , followed by a calibration function mapping R to saturation. The coefficients must be derived or verified through an appropriate calibration/validation procedure; hard-coding a generic formula is not sufficient for a commercial health claim.

## 17.4 Temperature Trend Model

Temperature processing should include a warm-up period, sensor calibration offset, moving median/low-pass filtering and an environmental/charging-context flag. Fever-related insight should be based on deviations from the user's baseline plus configurable absolute thresholds, and should emphasize "possible elevated temperature pattern" unless a validated clinical claim has been achieved.

## 17.5 ECG Signal Processing

The ECG path should implement validated high/low frequency filtering, 50 Hz mains rejection appropriate to the Nigerian power environment, lead-off detection, amplitude sanity checks and R-peak detection. A Pan-Tompkins-style or modern equivalent beat detector can be used as a starting point, but production implementation should be tested against annotated ECG datasets and a reference device.

## 17.6 AFib Research Pipeline

57. Acquire sufficiently clean single-lead ECG segments with signal-quality gating.
58. Detect R-peaks and construct RR interval series.
59. Extract rhythm-regularity, entropy, Poincare and morphology-related features where defensible.
60. Build a baseline rule-based classifier for triage and a supervised ML model as a research comparator.
61. Use patient-level train/validation/test separation to prevent subject leakage.
62. Evaluate sensitivity, specificity, PPV, NPV, AUROC, AUPRC and indeterminate-record rate.
63. Validate prospectively against a clinical ECG reference before any diagnostic marketing claim.

## 17.7 Sleep Apnea Research Pipeline

The MVP can investigate sleep-related patterns using PPG-derived pulse rate variability, SpO2 desaturation patterns and motion/stillness. However, because the baseline device lacks a direct airflow or respiratory-effort sensor, the initial feature should be framed as “sleep breathing risk signal” or “sleep wellness insight” during R&D. If strong evidence requires additional channels, the hardware should be expanded rather than forcing an under-specified model into production.

## 17.8 Fatigue Monitoring

The fatigue insight should use a personalized baseline rather than a universal threshold. Candidate features include resting HR trend, RMSSD/other HRV indices from validated ECG segments, activity load from accelerometry, sleep duration proxy from user/device state and temperature deviation. A simple score can be used during early development, followed by a trained model once sufficient longitudinal data and labels are available.

## 17.9 Model Governance

| Artifact         | Required control                                                  |
|------------------|-------------------------------------------------------------------|
| Dataset          | Versioned; provenance; inclusion/exclusion criteria               |
| Feature pipeline | Versioned code + deterministic preprocessing                      |
| Model            | Unique model ID; training date; framework version                 |
| Validation       | Locked test set; subgroup analysis; confidence intervals          |
| Deployment       | Model version stored with every result                            |
| Rollback         | Ability to disable a model and revert to previous version         |
| Monitoring       | Drift, missingness, quality distribution, false-positive feedback |

# 18. Web Platform and Backend Design

## 18.1 React Frontend

The public site should communicate the product, technical positioning, pricing, purchase path, support material and trust signals. The protected admin interface should expose order states, customer support tools, device/service telemetry and operational configuration. React provides reusable components and a clear separation between storefront UX and admin workflows.

## 18.2 FastAPI Services

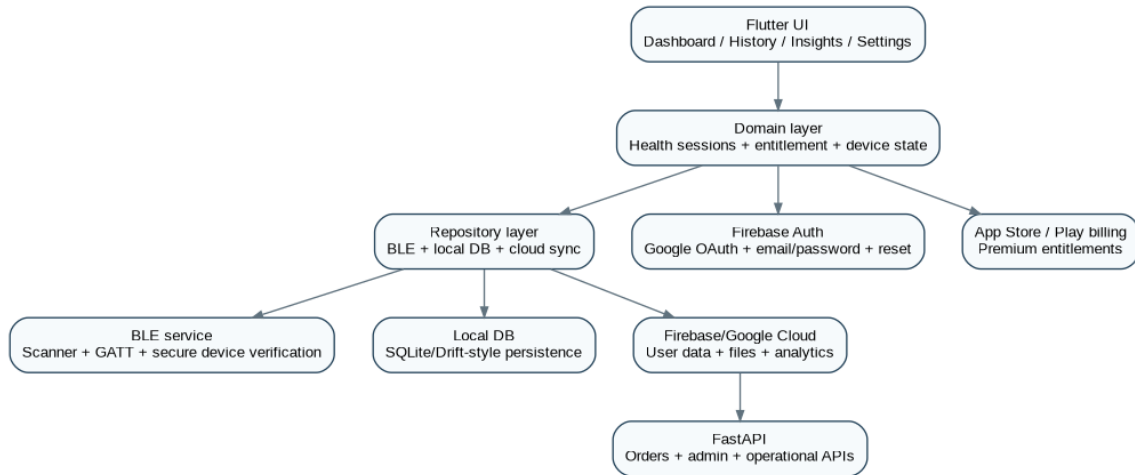


Figure 7. Software service boundaries; FastAPI provides operational and integration APIs.

| Service              | Responsibilities                                | Primary data                |
|----------------------|-------------------------------------------------|-----------------------------|
| Identity adapter     | Verify Firebase ID tokens and map user identity | Firebase Auth               |
| Device service       | Device metadata, activation, diagnostic state   | PostgreSQL/Firestore        |
| Order service        | Cart/order/payment status                       | PostgreSQL                  |
| Fulfillment service  | Delivery stages, tracking IDs                   | PostgreSQL                  |
| Insight service      | Insight metadata, model versions, job requests  | Cloud + PostgreSQL metadata |
| Admin service        | RBAC, audit logs, configuration                 | PostgreSQL                  |
| Telemetry admin      | Aggregate device health and firmware stats      | Analytics/telemetry store   |
| Notification service | Transactional messages and push triggers        | FCM/email provider          |

## 18.3 Render Deployment

Render can host the React and FastAPI services, with environment variables for secrets and a CI/CD pipeline connected to version control. FastAPI's official deployment documentation emphasizes production process management, HTTPS and deployment strategies; its security tools support OAuth2 and related authentication patterns. [26][27] Health checks, structured logs and an explicit rollback path should be configured before production launch.

# 19. Cybersecurity and Privacy Architecture

## 19.1 Threat Model

| ID | Threat                                | Impact                            | Primary control                                                                 |
|----|---------------------------------------|-----------------------------------|---------------------------------------------------------------------------------|
| T1 | Unauthorized BLE device impersonation | False telemetry / privacy leakage | Mutual verification, secure provisioning, authenticated session                 |
| T2 | Mobile token theft                    | Account compromise                | Firebase-managed auth, secure storage, short-lived tokens, device protections   |
| T3 | Local data extraction                 | Health-data disclosure            | OS-secure storage, encrypted DB/file where appropriate, minimal local retention |
| T4 | Cloud misconfiguration                | Mass data disclosure              | Least-privilege IAM, Firestore rules, environment separation, monitoring        |
| T5 | Admin privilege abuse                 | Mass modification/access          | RBAC + MFA where available +                                                    |

| ID | Threat                  | Impact                 | Primary control                                                                                          |
|----|-------------------------|------------------------|----------------------------------------------------------------------------------------------------------|
| T6 | Firmware tampering      | Device compromise      | audit logs + separation of duties<br>Signed firmware, secure boot/flash encryption strategy as supported |
| T7 | OTA supply-chain attack | Large fleet compromise | Signed releases + staged rollout + rollback                                                              |
| T8 | Algorithm drift         | Incorrect insight      | Model versioning + monitoring + disable/rollback controls                                                |
| T9 | Battery/charging fault  | Thermal/fire hazard    | Hardware protection + qualified battery + production test                                                |

## 19.2 Privacy-by-Design

The Nigeria Data Protection Act 2023 and NDPC guidance make data governance a core product concern. NDPC emphasizes data minimization, lawful and transparent processing, retention limits and appropriate security; its materials also identify privacy-by-design as a mandatory design principle and describe DPIA considerations for high-risk processing. [28][29][30] Because Guardian Watch processes physiological information, the project should treat privacy impact assessment and health-data governance as design gates, not legal paperwork added after development.

- Collect only the user information necessary for the requested service.
- Keep raw ECG and high-frequency data local by default where feasible, and make cloud upload user-controlled or explicitly governed.
- Use pseudonymous device identifiers internally; avoid using email addresses in device telemetry records.
- Provide user-visible export, deletion and unlinking controls.
- Log administrative access to health-related records.
- Develop a formal incident-response and breach-notification procedure before commercial launch.

## 19.3 Mobile Security Baseline

OWASP's Mobile Application Security Verification Standard provides a practical baseline across storage, cryptography, authentication, network security, platform interaction, code quality, resilience and privacy. [31] Guardian Watch should use MASVS as a verification checklist in addition to Firebase controls.

# 20. User Experience and Human Factors

## 20.1 Design Principles

- One primary action per screen during health sessions.
- Always show whether the device is connected, disconnected or synchronizing.
- Never present a low-quality physiological reading without a quality indication when the quality status materially affects interpretation.
- Use plain-language explanations for insights and distinguish “measurement” from “interpretation.”
- For ECG, provide electrode placement diagrams and a contact-quality indicator before recording.
- Allow dark/light themes and maintain legible typography at small watch sizes.
- Do not rely on color alone for safe/unsafe status.

## 20.2 Usability Engineering

IEC 62366-1 describes a usability-engineering process intended to identify and mitigate use-related risks for medical devices. [32] Even if the first Guardian Watch release remains within a consumer-wellness scope, the same discipline is valuable for onboarding, ECG placement, charging, device pairing, alert interpretation and data deletion.

## 21. Testing and Validation Strategy

### 21.1 Test Pyramid

| Level             | Examples                                                | Exit criterion                                          |
|-------------------|---------------------------------------------------------|---------------------------------------------------------|
| Unit              | Drivers, signal processing, repositories, parsers       | Automated tests pass; target coverage agreed per module |
| Integration       | BLE + mobile DB + cloud sync                            | End-to-end happy and failure paths pass                 |
| Hardware bring-up | Power rails, I2C/SPI, ADC, display, charging            | All rails within tolerance; peripherals enumerate       |
| Signal validation | PPG/ECG/temp against references                         | Performance meets pre-defined engineering targets       |
| Security          | Auth, BLE, API, role checks, mobile storage             | No critical/high findings unresolved                    |
| Usability         | Setup, ECG session, history, alerts                     | Critical tasks completed without unsafe confusion       |
| Production        | ICT/functional fixture, charging, display, sensor check | Yield and defect thresholds met                         |

### 21.2 Proposed Engineering Targets

| Metric              | Pre-production target                                                                   | Reference/standardization note                                 |
|---------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Heart-rate rest     | MAE $\leq$ 5 bpm target                                                                 | Compare with validated ECG/chest strap; not a regulatory claim |
| Heart-rate activity | MAE $\leq$ 7 bpm target for selected activities                                         | Evaluate walking, running, cycling, resistance separately      |
| SpO2                | Bias target within approximately $\pm 2$ percentage points in validated rest conditions | Requires calibrated reference and defined conditions           |
| Temperature         | After calibration, target $\leq 0.2$ C absolute error around normal range               | Location-dependent; validate against reference thermometer     |
| ECG                 | Stable waveform with low-lead-off and sample-loss rate                                  | Validate signal integrity against reference ECG                |
| BLE telemetry       | $< 1\%$ packet loss after retransmission/sequence recovery in nominal range             | Test distance/interference profiles                            |
| Battery             | At least one-day target for representative mixed use                                    | Measure with display duty cycle and real sensor profiles       |

**VALIDATION TARGETS:** These are engineering acceptance targets proposed for development, not established clinical-performance claims. Final thresholds must be locked in the Verification & Validation Plan after feasibility testing and regulatory assessment.

### 21.3 Reference Cohorts

Because published reviews highlight subgroup and context effects, validation should recruit a diverse cohort spanning sex, age, skin tone, wrist circumference, activity level and common wearing styles. Testing should include rest, walking, running, cycling, resistance training, varying strap tightness, ambient temperature and different device positions. [10][11]

### 21.4 Statistical Reporting

- Mean error and mean absolute error
- Bias with 95% limits of agreement where appropriate
- Sensitivity, specificity and predictive values for classification tasks
- AUROC/AUPRC for imbalanced classification
- Indeterminate/low-quality recording rate
- Subgroup performance and fairness analysis
- Confidence intervals and participant-level resampling

## 22. Manufacturing and Production Engineering

### 22.1 PCBWay Manufacturing Path

PCBWay is selected as the planned PCB fabrication/assembly partner. The production package should include Gerbers, drill files, pick-and-place data, BOM, assembly drawing, controlled component substitutions and a test specification. The design should be released only after ERC/DRC, footprint verification and 3D mechanical interference checks.

### 22.2 Prototype Phases

64. EVT (Engineering Validation Test): prove electrical architecture and sensor acquisition.
65. DVT (Design Validation Test): validate mechanical enclosure, power endurance, sensor performance, BLE stability and user experience.
66. PVT (Production Validation Test): run pilot production with manufacturing fixtures, yield analysis, labeling and packaging.
67. Mass production: release only after design freeze, certification gates and controlled change management.

### 22.3 Factory Test Fixture

- USB power input and charging-state test.
- 3.3 V / 1.8 V / 5 V rail verification.
- ESP32 programming and secure identity provisioning.
- Display pixel/area test.
- Buzzer electrical/acoustic test.
- I2C/SPI sensor enumeration.
- PPG optical self-test/known-reflector check.
- ECG analog loopback or simulator test.
- BLE advertising and connection test.
- Battery gauge readback and charge-cycle spot test.

### 22.4 Mechanical Production

The enclosure should move from 3D-printed prototypes to injection-molded or equivalent production tooling only after the PCB and component stack are frozen. The strap should be selected for repeated sweat exposure, cleaning and skin contact. The display lens should be optically compatible with the selected circular panel and should not create a large air gap over the optical sensing region.

### 22.5 Packaging

- Product box + internal tray
- Watch unit
- Strap
- USB Type-C cable/charger accessory as defined by SKU
- ECG cable/electrodes if included in the package
- Quick-start guide
- Full user manual / QR code to digital manual
- Warranty and support information
- Regulatory/product labels and serial/QR identification

## 23. Complete Bill of Materials

**BOM CONTROL:** This BOM is an engineering baseline for the wearable and its production ecosystem. Final manufacturer part numbers, approved substitutes, package variants and quantities must be frozen in an AVL before a production purchase order. Values marked “design-dependent” should be resolved by schematic simulation/reference-design review rather than blindly copied.

| Ref | Part / baseline                  | Qty | Function                          | Suggested vendor   | Engineering note                                |
|-----|----------------------------------|-----|-----------------------------------|--------------------|-------------------------------------------------|
| U1  | ESP32-C3-WROOM-02                | 1   | Core MCU + BLE/Wi-Fi              | Espressif          | Main processor; exact flash SKU to be frozen    |
| U2  | MAX30101                         | 1   | PPG / pulse / SpO2 optical sensor | Analog Devices     | 1.8 V logic + separate LED rail                 |
| U3  | MAX30205MTA+                     | 1   | Temperature sensor                | Analog Devices     | Skin-contact thermal location                   |
| U4  | AD8232ACPZ-R7                    | 1   | Single-lead ECG front end         | Analog Devices     | 20-pin LFCSP                                    |
| U5  | ADS1115IDGS                      | 1   | External precision ADC            | Texas Instruments  | ECG digitization; verify rate/latency           |
| U6  | LIS2DW12TR                       | 1   | 3-axis accelerometer              | STMicroelectronics | Motion + activity + artifact context            |
| U7  | MAX17048                         | 1   | Li-ion fuel gauge                 | Analog Devices     | Battery SOC estimation                          |
| U8  | W25Q64JV                         | 1   | 64-Mbit / 8 MB SPI NOR flash      | Winbond            | Local telemetry/ECG buffer                      |
| U9  | PCA9306                          | 1   | I2C level translator              | Texas Instruments  | 3.3 V ↔ 1.8 V MAX30101 bus                      |
| U10 | TPS7A2018-class 1.8 V LDO        | 1   | Low-noise 1.8 V rail              | Texas Instruments  | Exact part/package to be frozen                 |
| U11 | TPS63031-class 3.3 V buck-boost  | 1   | Main 3.3 V regulation             | Texas Instruments  | Battery-to-3V3 rail                             |
| U12 | 5 V boost regulator              | 1   | MAX30101 LED supply               | TI/ADI equivalent  | Must support transient LED current              |
| U13 | BQ24074-class charger/power path | 1   | Li-ion charger + power path       | Texas Instruments  | Preferred over bare TP4056 board for production |
| U14 | USB ESD protector                | 1   | ESD protection                    | TI/ST/Nexperia     | Place at USB connector                          |
| BZ1 | 3.3 V SMD piezo/magnetic buzzer  | 1   | Audible feedback                  | Approved vendor    | Current/polarity per selected part              |
| D1  | Power/status LED                 | 1   | Manufacturing/status indicator    | Approved vendor    | Prefer low-current LED                          |

| Refs    | Value / baseline                            | Qty | Function                                 | Suggested vendor | Engineering note                             |
|---------|---------------------------------------------|-----|------------------------------------------|------------------|----------------------------------------------|
| R1-R2   | 5.1 kOhm 1% 0402/0603                       | 2   | USB Type-C CC pull-downs                 | Yageo/Murata     | CC1/CC2 sink configuration                   |
| R3-R8   | 4.7 kOhm 1%                                 | 6   | I2C pull-ups / control pulls             | Yageo            | Only populate where bus topology requires    |
| R9-R20  | 100 Ohm-1 MOhm, 1% mixed                    | 12  | ECG gain/filter/lead-off network         | Yageo            | Exact values from final AD8232 filter design |
| R21-R28 | 10 kOhm-100 kOhm, 1%                        | 8   | Button, reset, ADC and control bias      | Yageo            | Exact values design-dependent                |
| R29-R32 | LED/backlight current-setting values        | 4   | Backlight / status LED                   | Yageo            | Exact values per display module              |
| C1-C12  | 100 nF X7R 16/25 V                          | 12  | IC high-frequency decoupling             | Murata/Samsung   | At/near IC power pins                        |
| C13-C20 | 1 uF X7R 10 V                               | 8   | Local rail decoupling                    | Murata/Samsung   | Power IC/sensor local decoupling             |
| C21-C26 | 4.7-22 uF low-ESR ceramic                   | 6   | Bulk rail support                        | Murata/Samsung   | Exact values per regulator                   |
| C27-C36 | 10 pF-100 nF mixed                          | 10  | ECG filters / analog signal conditioning | Murata           | Exact values from validated filter topology  |
| L1      | Inductor per 3V3 regulator reference design | 1   | 3.3 V buck-boost energy storage          | Murata/Coilcraft | Must follow regulator reference design       |
| L2      | Inductor per 5V boost reference design      | 1   | 5 V optical LED rail                     | Murata/Coilcraft | Select for peak LED current                  |
| L3      | Ferrite bead 600 Ohm class @100 MHz         | 1   | Power/analog isolation                   | Murata           | Optional after EMI test                      |

| Ref | Part                                  | Qty | Function                    | Suggested vendor | Engineering note                      |
|-----|---------------------------------------|-----|-----------------------------|------------------|---------------------------------------|
| J1  | USB Type-C receptacle, USB2.0, 16-pin | 1   | Charging/programming access | GCT/HRO/JAE      | Mounting and shell tie-downs required |

| Ref | Part                                                 | Qty   | Function                 | Suggested vendor      | Engineering note                               |
|-----|------------------------------------------------------|-------|--------------------------|-----------------------|------------------------------------------------|
| J2  | 3.5 mm TRS/TRRS ECG jack or low-profile equivalent   | 1     | External ECG electrodes  | CUI/Same Sky/approved | Must be mechanically retained                  |
| J3  | Battery connector JST-PH 2-pin or locking equivalent | 1     | Battery connection       | JST                   | Select wire gauge for charge/discharge current |
| J4  | Programming/test pads                                | 1 set | Factory programming/UART | PCB pads              | Not user accessible                            |
| J5  | Display FPC/board-to-board connector                 | 1     | GC9A01 display           | Hirose/TE/appropriate | Depends on chosen display module               |
| J6  | Strap mechanical mounts                              | 2     | Wrist strap retention    | Mechanical            | Integrated into enclosure                      |

| Ref  | Part                                     | Qty | Function                       | Suggested vendor           | Engineering note                                  |
|------|------------------------------------------|-----|--------------------------------|----------------------------|---------------------------------------------------|
| BT1  | 3.7 V LiPo 500-600 mAh protected         | 1   | Primary energy source          | Qualified battery supplier | Require traceability, test summary and cycle data |
| SW1  | Tactile/slide power button               | 1   | User power control             | ALPS/CK                    | Long-press supported in firmware                  |
| F1   | Resettable fuse or input protection fuse | 1   | USB input protection           | Littelfuse                 | Value based on charger/input budget               |
| NTC1 | 10 kOhm NTC (battery thermistor)         | 1   | Battery temperature monitoring | Murata/TDK                 | Required if charger uses NTC input                |
| TP1  | Battery test pads                        | 1   | Factory test                   | PCB                        | Accessible on fixture                             |

| Ref   | Part                                       | Qty | Function                    | Suggested vendor            | Engineering note                                        |
|-------|--------------------------------------------|-----|-----------------------------|-----------------------------|---------------------------------------------------------|
| DISP1 | 1.28 in circular TFT, 240x240, GC9A01, SPI | 1   | User interface              | Approved display house      | Prefer module with documented FPC and backlight current |
| LENS1 | Optical-grade clear lens/cover             | 1   | Display/surface protection  | Injection-mold supplier     | Hard coat and scratch resistance target                 |
| GASK1 | Silicone/TPU sealing gasket                | 1   | Environmental protection    | Approved elastomer supplier | Geometry after enclosure freeze                         |
| OPT1  | Black optical isolation gasket/foam        | 1   | PPG ambient-light isolation | Optical/mechanical supplier | Avoid direct light path leakage                         |

| Ref   | Part                            | Qty    | Function                         | Supplier class                 | Engineering note                                 |
|-------|---------------------------------|--------|----------------------------------|--------------------------------|--------------------------------------------------|
| MECH1 | Upper chassis shell             | 1      | Structural enclosure             | Injection-molding supplier     | PC/ABS/PA or selected polymer                    |
| MECH2 | Bottom sensor housing           | 1      | Sensor/skin interface            | Injection-molding supplier     | Use skin-compatible material at contact surfaces |
| MECH3 | Watch strap, quick-release      | 1 pair | Wearable retention               | Silicone/TPU supplier          | Sweat/tear-resistant                             |
| MECH4 | Lens adhesive / gasket adhesive | 1 set  | Display sealing                  | 3M/medical-compatible supplier | Validate biocompatibility where applicable       |
| MECH5 | M1.4/M1.6 stainless screws      | 4-8    | Enclosure assembly               | Approved hardware              | Quantity depends on enclosure design             |
| MECH6 | Threaded inserts if required    | 4      | Structural screw retention       | Brass/stainless                | Optional based on molded chassis                 |
| MECH7 | ECG jack dust cover/cap         | 1      | Protection when jack unused      | Silicone/TPU                   | Recommended                                      |
| MECH8 | USB port gasket                 | 1      | Ingress protection around Type-C | Elastomer supplier             | Validate plug/cable compatibility                |
| MECH9 | Adhesive foam / spacers         | 1 set  | PCB positioning                  | 3M/tesa                        | Compression-controlled                           |

| Ref  | Item                          | Qty        | Function              | Supplier           | Engineering note               |
|------|-------------------------------|------------|-----------------------|--------------------|--------------------------------|
| ASM1 | Bare 4-layer PCB, ENIG finish | 1          | Wearable main board   | PCBWay             | Dimensions fixed after routing |
| ASM2 | SMT assembly service          | 1          | PCB population        | PCBWay             | Include AOI where available    |
| ASM3 | Firmware provisioning         | 1          | Factory software load | In-house/contract  | Secure provisioning station    |
| ASM4 | Device QR/serial label        | 1          | Traceability          | Printing supplier  | Unique device ID               |
| ASM5 | Functional test fixture       | 1 per line | Production test       | In-house           | Pogo pins + BLE + USB test     |
| ASM6 | Packaging tray                | 1          | Product protection    | Packaging supplier | Thermoformed/ molded           |
| ASM7 | Retail carton                 | 1          | Brand packaging       | Packaging supplier | Include regulatory labels      |
| ASM8 | User manual/quick-start       | 1          | Documentation         | Print supplier     | QR to full digital manual      |
| ASM9 | Warranty card                 | 1          | After-sales           | Print supplier     | Serial/activation reference    |



## 23.1 PCB Component Population Summary

| Category               | Approx. quantity | Notes                                                |
|------------------------|------------------|------------------------------------------------------|
| ICs / modules          | 14-16            | Core MCU, sensors, converters, charger, ADC, storage |
| Resistors              | 50+              | Values finalized by schematic and reference designs  |
| Capacitors             | 30+              | Decoupling, regulators, filters                      |
| Inductors / ferrites   | 3-5              | Power conversion and EMI                             |
| Connectors             | 4-6              | USB-C, ECG, battery, display, test                   |
| Mechanical parts       | 8-12             | Chassis, lens, gasket, strap, fasteners              |
| Production consumables | variable         | Solder paste, labels, adhesives, packaging           |

**BOM RELEASE DISCIPLINE:** “Complete BOM” at project-report stage means every functional assembly element and its procurement category is represented. The final released BOM should additionally contain manufacturer part number, manufacturer, approved substitute, package/footprint, lifecycle status, unit cost, MOQ, source, AVL status and revision-control metadata for each line item.

## 24. Software and Cloud Bill of Materials

| Layer    | Component                              | Purpose                          | License/operational note                             |
|----------|----------------------------------------|----------------------------------|------------------------------------------------------|
| Firmware | ESP-IDF / C/C++ toolchain              | ESP32 application development    | Pin version in repository                            |
| Firmware | FreeRTOS facilities in ESP-IDF         | Tasks/timers/synchronization     | Use deterministic task boundaries                    |
| Firmware | BLE GATT stack in ESP-IDF              | Wireless transport               | Version and test protocol                            |
| Firmware | LittleFS or equivalent                 | Local flash filesystem           | If filesystem used; ring buffer logic still required |
| Mobile   | Flutter/Dart                           | Android/iOS UI and app logic     | Pin SDK and plugin versions in CI                    |
| Mobile   | Firebase Core/Auth libraries           | Cloud bootstrapping and identity | Google/Apple platform setup required                 |
| Mobile   | Google Sign-In package                 | OAuth provider flow              | Use current maintained package                       |
| Mobile   | BLE package                            | Scanning, GATT and connection    | Evaluate native compatibility per OS release         |
| Mobile   | SQLite/Drift-style local DB            | Offline health data              | Encrypt sensitive data where needed                  |
| Mobile   | In-app purchase package                | Premium entitlements             | StoreKit + Google Play Billing integration           |
| Mobile   | Charts/visualization                   | History trends                   | Avoid rendering raw ECG at unsafe/incorrect scales   |
| Cloud    | Firebase Authentication                | Identity                         | Managed credentials                                  |
| Cloud    | Cloud Firestore / selected store       | User/app data                    | Use strict security rules                            |
| Cloud    | Cloud Storage                          | ECG/file objects                 | Security rules + signed access                       |
| Cloud    | Firebase App Check (where appropriate) | Abuse/app authenticity signal    | Do not treat as authorization                        |
| Cloud    | Crash reporting/monitoring             | Reliability                      | PII-aware logging                                    |
| Backend  | Python + FastAPI                       | Operational API                  | Pin Python and dependency versions                   |
| Backend  | Pydantic                               | Validation/schema                | API contract                                         |
| Backend  | Uvicorn/Gunicorn-style process         | Server runtime                   | Configure health checks and workers                  |
| Backend  | PostgreSQL                             | Orders/admin/audit metadata      | Managed DB with backups                              |
| Web      | React + TypeScript                     | Storefront/admin                 | Use a current supported Node release                 |
| Web      | Router/state/query libraries           | Navigation and data fetching     | Architecture-dependent                               |
| DevOps   | Git + CI/CD                            | Source control/build             | Protected main branch and signed releases            |
| DevOps   | Render                                 | Hosting for web/API services     | Secrets via environment variables                    |
| Ops      | Google Cloud Console                   | Cloud monitoring/configuration   | Least-privilege IAM                                  |
| Security | OWASP MASVS baseline                   | Mobile security verification     | Use as test checklist                                |
| Docs     | OpenAPI/Swagger generated by FastAPI   | API contract                     | Version API schemas                                  |

## 24.1 Environment Separation

| Environment | Purpose                | Required controls                                    |
|-------------|------------------------|------------------------------------------------------|
| DEV         | Developer work         | Synthetic/test users; no production secrets          |
| STAGING     | Integrated QA          | Production-like configuration; synthetic health data |
| PILOT       | Controlled field trial | Approved participants; monitored deployment          |
| PRODUCTION  | Commercial users       | Strict IAM, backups, monitoring, change control      |

## 25. Regulatory, Standards and Compliance Roadmap

### 25.1 Nigerian Regulatory Position

NAFDAC's current medical-device registration guidance (effective 3 May 2024) covers imported and locally manufactured medical devices. A current NAFDAC draft regulation groups medical devices by risk into Classes A through D, from low risk to high risk. [33][34] The product's actual classification must be determined from intended purpose, claims, functionality and applicable rules—not simply from the presence of ECG or medical-grade sensors.

SON's MANCAP programme is relevant to locally manufactured products and is intended to ensure conformity with applicable Nigerian Industrial Standards before market introduction. SON identifies electrical/electronics among the product sectors and describes inspection, sampling, testing, review and surveillance activities. [35][36]

The Nigeria Data Protection Act 2023 and NDPC oversight are also directly relevant because Guardian Watch handles personal and physiological data. [28][29]

### 25.2 International Standards to Evaluate

| Standard/framework                      | Area                                                                | Guardian Watch application                                                                                                                    |
|-----------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| ISO 14971:2019                          | Medical device risk management                                      | Hazard analysis, risk controls, residual risk and lifecycle monitoring. ISO states this edition remains current after 2025 confirmation. [37] |
| IEC 62304:2006+A1:2015                  | Medical device software lifecycle                                   | Use if software becomes part of a medical-device regulatory scope. [38]                                                                       |
| IEC 62366-1:2015+A1:2020                | Usability engineering                                               | Use for safety-related usability and human factors. [32]                                                                                      |
| IEC 60601-1                             | Medical electrical equipment basic safety and essential performance | Evaluate applicability if the device is classified as medical electrical equipment. [39]                                                      |
| IEC 81001-5-1:2021                      | Health software cybersecurity lifecycle                             | Relevant to secure development of health software. [40]                                                                                       |
| IEC 62133-2 / battery requirements      | Portable rechargeable lithium battery safety                        | Evaluate for battery/cell compliance depending on product and cell configuration.                                                             |
| UN Manual Tests & Criteria Section 38.3 | Lithium battery transport testing                                   | Battery transport/test-summary requirements. [22]                                                                                             |
| Bluetooth SIG Qualification             | Bluetooth product qualification                                     | Interoperability and qualification process. [24]                                                                                              |
| Applicable EMC/RF requirements          | Wireless/electronic compliance                                      | Market-specific conformity assessment for radio and electronics                                                                               |

### 25.3 Regulatory Gated Release Plan

68. Freeze intended use and marketing claims.
69. Perform Nigerian regulatory classification assessment with qualified regulatory counsel/consultant.
70. Create product risk-management file and usability plan.
71. Run component compliance review and gather certificates/data sheets.
72. Perform electrical, EMC, battery and RF pre-compliance testing as applicable.
73. Complete algorithm verification and clinical validation for any medical claim.

74. Prepare SON/MANCAP and NAFDAC documentation where applicable.
75. Finalize labels, packaging, user manual, warnings and support procedures.
76. Complete post-market surveillance, complaint handling and recall plan.

## 26. Quality Management and Lifecycle Controls

### 26.1 Configuration Management

| Artifact      | Revision control                                                |
|---------------|-----------------------------------------------------------------|
| PCB           | Git/PLM revision, Gerbers linked to exact BOM revision          |
| BOM           | Manufacturer PN + AVL + approved substitutes + lifecycle status |
| Firmware      | Signed release; hardware revision compatibility matrix          |
| Mobile        | Pinned SDK/dependency lockfile and build artifact hash          |
| Backend       | API versioning + database migrations                            |
| Models        | Dataset, code, model and threshold version                      |
| Documentation | Controlled version + approval record                            |

### 26.2 Change Control

Every production change should receive a change request, impact assessment and verification plan.

Substituting a regulator, sensor, battery, display or enclosure material can change safety or measurement behavior and should therefore never be treated as a simple procurement swap without impact analysis.

### 26.3 Failure Investigation

- Use serial/device logs to reconstruct field events.
- Classify failures as hardware, firmware, app, backend, data or user/usage.
- Track corrective and preventive actions (CAPA).
- Feed returned-device evidence back into design, manufacturing and validation datasets.

## 27. Project Implementation Roadmap

| Phase                | Duration    | Key work                                                           | Deliverable                                                    |
|----------------------|-------------|--------------------------------------------------------------------|----------------------------------------------------------------|
| P0 Requirements      | 2-3 weeks   | Claims, architecture, UX, BOM draft                                | System Requirements + Product Requirements                     |
| P1 Hardware EVT      | 4-6 weeks   | Schematic, PCB, prototype assembly, firmware bring-up              | EVT hardware + basic telemetry                                 |
| P2 Software MVP      | 6-8 weeks   | Flutter auth/BLE/dashboard/history; FastAPI/React skeleton         | Integrated software MVP                                        |
| P3 DVT               | 6-10 weeks  | Enclosure, power optimization, sensor validation, security testing | DVT samples + V&V evidence                                     |
| P4 Insight R&D       | 8-16+ weeks | Feature engineering, datasets, model validation                    | Validated wellness models; clinical roadmap for medical claims |
| P5 PVT               | 6-8 weeks   | Factory fixture, packaging, labels, manufacturing process          | Pilot production lot                                           |
| P6 Regulatory/Launch | variable    | Certification, documentation, channel setup                        | Commercial release decision                                    |

### 27.1 Team Structure

| Role                   | Primary responsibilities                                      |
|------------------------|---------------------------------------------------------------|
| Systems/Technical Lead | Architecture, requirements, integration and release decisions |
| Embedded Engineer      | PCB/firmware/BLE/power                                        |
| Signal/ML Engineer     | PPG/ECG/temperature algorithms, datasets, model validation    |
| Flutter Engineer       | Mobile app and device connectivity                            |
| Backend Engineer       | FastAPI, cloud sync, data model, admin services               |
| Frontend Engineer      | React storefront/admin                                        |
| Cybersecurity Engineer | Threat model, mobile/cloud security, secure provisioning      |

| Role                    | Primary responsibilities                                        |
|-------------------------|-----------------------------------------------------------------|
| Mechanical Engineer     | Enclosure, strap, optical/thermal interface                     |
| QA/V&V Engineer         | Test planning, traceability, regression, performance validation |
| Regulatory/Quality Lead | QMS, risk file, SON/NAFDAC/standards coordination               |

## 28. Risk Register and Mitigation

| ID  | Risk                            | Impact | Likelihood | Mitigation                                                        |
|-----|---------------------------------|--------|------------|-------------------------------------------------------------------|
| R1  | PPG motion artifact             | High   | High       | Motion sensor + signal-quality gating + diverse validation        |
| R2  | Skin-tone performance variation | High   | Medium     | Diverse cohort; subgroup reporting; calibration/algorithm checks  |
| R3  | False medical reassurance       | High   | High       | Claim controls; warnings; validation; regulatory review           |
| R4  | Battery thermal event           | High   | Low/Med    | Protected battery + charger + thermal tests + factory screening   |
| R5  | BLE impersonation               | High   | Medium     | Authenticated pairing and secure provisioning                     |
| R6  | Cloud data exposure             | High   | Medium     | Least privilege + rules + audit + DPIA + penetration testing      |
| R7  | Poor battery life               | Medium | High       | Power profiling early; display duty cycling; efficient regulators |
| R8  | ECG noise                       | High   | Medium     | 4-layer PCB, analog partition, filter validation, shielded leads  |
| R9  | Supply-chain shortage           | Medium | Medium     | AVL with alternates and lifecycle monitoring                      |
| R10 | Manufacturing low yield         | Medium | Medium     | DFM review, AOI, test fixture, pilot build                        |
| R11 | Algorithm overfitting           | High   | Medium     | Subject-independent split and locked external validation          |
| R12 | Regulatory rework               | High   | Medium     | Claims/classification gate before design freeze                   |

## 29. Cost and Commercialization Framework

A reliable unit cost model must include more than component cost. For Guardian Watch, the commercial model should separate hardware direct cost, manufacturing conversion cost, test/QA cost, packaging, logistics, warranty reserve, cloud/software cost, payment processing, regulatory/compliance expense and customer support.

| Cost bucket   | Examples                                    | Pricing model                                  |
|---------------|---------------------------------------------|------------------------------------------------|
| BOM           | ICs, passives, battery, display, connectors | Unit cost at pilot and 1k/10k volumes          |
| Manufacturing | PCB assembly, rework, testing               | Per-unit + NRE amortization                    |
| Mechanical    | Enclosure, strap, lens                      | Tooling amortization + unit                    |
| Packaging     | Box, manual, labels                         | Per unit                                       |
| Cloud         | Firebase/Google Cloud/Render                | Fixed + usage-based                            |
| Software      | Store fees, maintenance, support            | Per active user / transaction where applicable |
| Warranty      | Returns, RMA, replacement                   | Reserve percentage                             |
| Regulatory    | Testing, certification, consulting          | Project cost amortized over launch volume      |

### 29.1 Cost-Control Strategy

- Use a small set of high-value sensors and avoid redundant chips.

- Prefer components with multi-source availability and stable lifecycle.
- Keep the PCB 4-layer but compact enough for low panel cost.
- Use external flash only if local ECG buffering justifies it; it can be omitted from a lower-cost SKU.
- Offer multiple software SKUs rather than multiple hardware variants early.
- Use the same mobile/cloud stack for all hardware revisions through a versioned device abstraction.

## 30. Expected Results and Success Criteria

The first production-capable release should demonstrate end-to-end functionality rather than isolated component demos. The following outcomes define success:

77. Wearable powers safely, charges safely and achieves the target mixed-use runtime under measured test conditions.
78. All sensors initialize reliably and produce time-stamped data with quality metadata.
79. BLE onboarding is reliable and device identity verification prevents unauthorized association.
80. The mobile dashboard updates in real time, preserves records offline and synchronizes without duplicates.
81. History, ECG session and settings workflows work across supported Android/iOS test matrices.
82. Premium entitlement logic is correct for purchase, restore and cancellation states.
83. React storefront and admin functions support order-to-delivery state transitions.
84. Production test fixture identifies common hardware assembly defects before shipping.
85. Security testing finds no unresolved critical vulnerabilities.
86. All medical/health claims used in customer-facing materials are consistent with evidence and regulatory classification.

## 31. Limitations and Research Gaps

The proposed architecture is ambitious, and several capabilities require deliberate boundaries. First, a wrist-based PPG system is not equivalent to a clinical pulse oximeter or diagnostic ECG simply because similar sensor components are used. Second, wrist temperature has a different physical relationship to core body temperature and requires a validated protocol. Third, fatigue and sleep-apnea inference from wearable signals is an active research area with strong academic interest but variable real-world generalizability. [12][15][17][18] Fourth, the initial hardware does not contain a direct airflow/respiratory-effort channel, which constrains sleep-apnea inference.

Another limitation is the dependency on user behavior. Strap tightness, placement, movement, charging habits, electrode placement and app permissions can all affect data quality. The product should therefore measure and communicate signal quality rather than assuming perfect conditions.

## 32. Future Expansion

### 32.1 Hardware Roadmap

- Add a more integrated medical-grade biopotential AFE if ECG performance/validation demands it.
- Add gyroscope or upgraded IMU if activity/sleep classification requires richer motion features.
- Add dedicated respiratory/pressure sensing for stronger sleep-apnea research.
- Consider NFC for fast accessory/device pairing in later hardware.
- Evaluate wireless charging once enclosure and efficiency targets justify it.
- Introduce external memory capacity options for research/clinical sessions.

### 32.2 Software Roadmap

- Personalized baselines and longitudinal anomaly detection.

- Advanced recovery and training-load analytics.
- Clinician portal under a separately regulated product architecture if needed.
- Interoperability with healthcare data formats where legally and commercially appropriate.
- On-device TinyML for selected low-risk signal-quality tasks.
- Fleet device management, signed OTA updates and automated manufacturing telemetry.

### 33. Conclusion

Guardian Watch is technically feasible as a wearable fitness and wellness platform, provided the design is approached as a full system rather than as a collection of hobby electronics modules. The proposed architecture combines a capable low-power BLE MCU, optical and electrical biosensing, temperature and motion sensing, local storage, a deliberate multi-rail power system, an offline-first mobile application, cloud synchronization, operational APIs and a web commerce/administration platform.

The literature supports the value of wearable heart-rate and physiological monitoring but repeatedly shows that real-world measurement depends on context, motion, sensor placement, skin characteristics, signal quality and validation methodology. [10][11][12] Guardian Watch therefore treats signal quality as a first-class data attribute and treats advanced medical insights as staged research/validation products. AFib and sleep-apnea features should proceed through dedicated algorithmic and clinical validation rather than being released as unsupported medical claims. [13][14][15]

For Nigerian commercialization, the project should align manufacturing and product certification planning with SON MANCAP requirements where applicable, work through NAFDAC classification/registration where the product falls within medical-device scope, and implement privacy-by-design under the Nigeria Data Protection Act framework. [33][35][28] The technical baseline documented here is suitable for progressing to schematic capture, KiCad PCB design, firmware repository setup, Flutter app development, cloud environment creation, manufacturing RFQ and a structured verification-and-validation program.

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## Appendix A — Suggested Data Packet Format

```
TelemetryPacket {
  protocol_version: uint8
  device_id: bytes[8]
  sequence: uint32
  timestamp_ms: uint64
  flags: uint16
  hr_bpm_x10: uint16
  spo2_x10: uint16
  temp_c_x100: int16
  motion_rms_x1000: uint16
  quality_score: uint8
  crc16: uint16
}
```

Production protocol fields should be versioned and backward-compatible where possible. Large ECG packets should use separate sequence-framed characteristics or chunked object synchronization rather than forcing raw ECG into small notification packets.

## Appendix B — Example BLE State Machine

| State                | Entry                   | Exit              | Failure handling                 |
|----------------------|-------------------------|-------------------|----------------------------------|
| Disconnected         | No link                 | Scan/connect      | Backoff and user-visible status  |
| Connected-Unverified | BLE link established    | Authenticated     | Disconnect on failed auth        |
| Authenticated        | Handshake succeeds      | User session ends | Rotate session state if required |
| Streaming            | Telemetry enabled       | Pause/stop        | Sequence recovery                |
| Syncing              | Buffered data exists    | Offset complete   | Retry queue                      |
| Fault                | Hardware/software fault | Reset/recovery    | Log error code + safe shutdown   |

## Appendix C — Minimum Production Documentation Set

- Product Requirements Document (PRD)
- System Requirements Specification (SRS)
- Hardware Design Specification
- Electrical schematic and PCB files
- BOM + Approved Vendor List
- Firmware Design Specification
- BLE GATT specification
- Mobile Software Architecture Specification
- Backend API specification
- Database schema
- Cybersecurity threat model
- Risk management file
- Verification and Validation Plan and Report
- Usability engineering file where applicable
- Manufacturing test specification
- Factory work instructions
- Packaging artwork and labeling specification
- User manual
- Warranty/RMA procedure
- Privacy policy and data retention schedule
- Regulatory submission package where applicable

## Appendix D — Design Decisions to Freeze Before PCB Release

| Decision      | Recommended baseline                        | Freeze trigger                                     |
|---------------|---------------------------------------------|----------------------------------------------------|
| MCU           | ESP32-C3-WROOM-02                           | Module SKU + memory + RF/antenna strategy verified |
| Display       | 240x240 GC9A01 circular TFT                 | Exact display vendor/module + mechanical CAD       |
| PPG           | MAX30101                                    | Optical stack + LED current + validation plan      |
| Temperature   | MAX30205                                    | Thermal path + calibration procedure               |
| ECG           | AD8232 + external ADC                       | Electrode/jack + analog filter values + V&V        |
| Motion        | LIS2DW12                                    | Activity and artifact model requirements           |
| Battery       | 3.7 V 500-600 mAh protected LiPo            | Mechanical fit + transport evidence                |
| Charging      | BQ24074-class power path                    | Package + thermal + USB current budget             |
| Local storage | 8 MB SPI NOR                                | Expected offline retention policy                  |
| Enclosure     | Sealed polymer chassis + silicone/TPU strap | DFM + environmental test target                    |
| Compliance    | SON/NAFDAC/NDPC evaluation                  | Intended use and market claims frozen              |